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# Investigations on the antioxidant and UV-blocking properties of biodegradable chitosan films enriched with MgO nanoparticles and red dragon fruit peel extract

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ABSTRACT

The aim of the study is to investigate the synergistic impact of MgO nanoparticles and red dragon fruit peel extract (RFE) on chitosan (CS) biofilm. The CS films were analyzed using FTIR, TGA, ESEM/EDX, XRD, UV-spectra analysis along with contact angle measurement, antioxidant activity, overall migration and biodegradation study. The results showed that as the concentration of RFE increases, improvement in UV-blocking and thermal properties is observed. The FTIR analysis revealed the hydrogen bond interactions among CS, MgO nanoparticles, and RFE. The findings showed that a significant ($P < 0.05$) increase in 2,2-Diphenyl-1-picrylhydrazyl (DPPH) and 2,2-azino-bis-3-ethylbenzothiazoline-6-sulphonic acid (ABTS) radical scavenging activity, reaching to 37.63 % and 62.83 %, respectively. Also, increase in moisture content, water solubility, and water vapor permeability up to 25.64 %, 38.58 %, and $2.83 \times 10^{-10}$ g $m^{-1}$ $Pa^{-1}$ $s^{-1}$ respectively, was found. ESEM images depicted the aggregation of MgO nanoparticles. The films comply with EU 10/2011 safety standards for fatty foods.

## 1. Introduction

Environmental problems caused by petroleum-based packing materials are very well known. The massive amount of plastic waste released into the environment on a global scale is of great concern. During the circulation cycle of waste plastic; after some period of time, plastic gets broken down into nanoplastic and microplastic (Shen et al., 2020). Microplastic and nanoplastic pose a severe threat to human health, which is well documented in the literature (Revel et al., 2018). The researchers' attention is therefore shifted towards developing the novel biodegradable film packaging materials as an alternative to plastic packaging (Yong et al., 2019).

Dragon fruit is most widely recognized as an exotic fruit. Based on the variety of color of the pulp, it can be distinctly classified as white, red, and yellow. Its peels, however, occupies around 30–35 % of its weight, is mostly neglecled as a byproduct and discarded. Polyphenolic compounds, betalain pigment compounds, and vitamins are usually present in the dragon fruit peels. In the previous study, different deep eutectic solvents were used to extract betacyanin compounds from red dragon fruit peel. The deep eutectic solvent extract, rich in phenolic compounds and betalain pigment compounds, was used as a plasticizer in chitosan (CS) based biofilm (Prajapati and Jadeja, 2023). Therefore, its peels can be utilized as a natural source for these value-added compounds.

The CS is made up of β-(1,4)-D-glucosamine and N—acetyl-D-glucosamine. It is biodegradable and non-toxic in nature. Also, it has good film-forming properties (Sanuja et al., 2014). However, it has limited applicability due to low water solubility and lack of anti-oxidant properties. To solve this problem and widen the horizon of application of CS film for various purposes, the approach of the addition of nanomaterials and various plant extracts is used (Liu et al., 2021b). The literature reports varying combinations of different nanomaterials and different plant extracts/essential oil. Purple corn extract and silver nanoparticles showed to improve antioxidant property, and mechanical property (Qin et al., 2019). Incorporating titanium dioxide and litchi peel extract into CS film increased the shelf life of apples (Liu et al., 2021a). Similarly, the synergistic incorporation of bamboo leaves and ZnO nanoparticles improved the antibacterial properties (Liu et al., 2021b).

MgO nanoparticles are non-toxic and cheaper (De Silva et al., 2017). MgO nanoparticles represent a ubiquitously utilized metal oxide synthesized from natural minerals such as dolomite and magnesite.

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Creating an alkaline environment with an aqueous $Mg(OH)_2$ solution in water can effectively combat a wide range of fungi, pathogenic bacteria, and viruses even at low concentrations (Cai et al., 2018). MgO nanoparticles have garnered the designation of "generally recognized as safe" from the US Food and Drug Administration (USFDA). Therefore, MgO nanoparticles can be used in a variety of applications without compromising the standards of safety and regulatory compliance (Luo et al., 2022). During the literature review, few studies were found in which a synergistic effect MgO nanoparticles and plant extract on biopolymers was investigated. Few studies were found in which MgO nanoparticles and essential oil were added to the CS film. The addition of clove oil (Sanuja et al., 2014) and *Ziziphora clinopodioides* essential oil (Naeeji et al., 2020) with MgO nanoparticles was reported. 5 wt% and 10 wt% of MgO nanoparticles were added to CS film, and significant improvement in thermal, mechanical, and barrier properties was found. In this study, information regarding the effect of the addition of any plant/fruit peel extract in combination with MgO nanoparticles was lacking (De Silva et al., 2017). The introduction of an optimum concentration of MgO nanoparticles along with tea polyphenols improved the tensile strength, hydrophobicity, and thermal stability of the potato starch based biofilms. Concurrently, a decrease in moisture content, water solubility, and water vapor permeability was reported (Luo et al., 2022).

No research work was found in which a combination of red dragon fruit peel lyophilized extract powder and MgO nanoparticles were incorporated to CS film. Therefore, in the current study, CS film was fabricated with the addition of different concentrations of freeze-dried dragon fruit peel extract powder and MgO nanoparticles. Furthermore, various analytical characterizations of CS blended films, such as FTIR, XRD, TGA, contact angle, mechanical properties, etc., were carried out. In addition, the study of water solubility, water vapor permeability, moisture content, opacity, transparency, and UV-blocking ability of the CS films was carried out. The present study hypothesizes that adding nanoparticles and peel extract should improve the properties of CS films, such as antioxidant property, thermal stability, light properties, etc.

## 2. Materials and methods

CS (Medium molecular weight, 90 % degree of deacetylation and 150–500 m.Pas viscosity at 20 °C) was purchased from Loba Chemie, Mumbai, India. Red dragon fruits were obtained from the local market in Surat, India. Glycerol was purchased from Finar Ltd., India. Magnesium Oxide nanoparticles, DPPH, and ABTS were obtained from Sigma Aldrich, India.

### 2.1. Preparation and extraction of red dragon fruit peel powder

Red dragon fruit peel powder was prepared using the method previously described, and extraction was carried out using the optimum conditions obtained in previous work using water as solvent (Prajapati and Jadeja, 2023). The water extract was kept in the deep freezer (Blue star Ltd., CF5-460NEYW, India) at −20 °C for overnight. Then, the freeze drying (Freeze drying systems Pvt. Ltd., India) was carried out at −40 °C temperature for 8 h. Finally, the red dragon fruit peel extract powder (RFE) was obtained, which was utilized to prepare the CS based biofilm.

### 2.2. Characterization of the RFE

0.25 g of RFE was dissolved in 25 mL water and kept in a stirrer (Remi lab world, 5 MLH plus, India) until a homogenous solution was obtained. The resulting solution was then sent for the quantification of total phenolics, total betacyanin, and betaxanthin content. The measurement of the total phenolics (Mahayothee et al., 2019), total betacyanins, and total betaxanthin content (Sharma et al., 2021) was carried out using the method described in literature.

### 2.3. CS based biofilm preparation method

0.50 g of CS was dissolved in 1 % acetic acid solution (50 mL) and kept in magnetic stirring (Remi lab world, 5 MLH plus, India) overnight at 350 rpm. After that, 5 % (*w/v*) MgO nanoparticles were added to the film solution. The concentration of MgO nanoparticles was chosen based on a previous study (De Silva et al., 2017). The preweighed RFE was added to the solution, and then it was subjected to sonication for 10 min in an ultrasonic processor (Hielscher Ultrasonics, UP200Ht, Germany) and a titanium alloy probe (7 mm diameter). Afterwards, the solution was kept in stirring for 10 min to achieve a homogenous solution. Then, 25 mL of film forming solution was poured into a 25 cm diameter acrylic plate and dried at room temperature. Finally, the films were then peeled off from the acrylic plate and kept in a polyethylene bag until further use. Based on the composition dictated in Table S1, different concentrations of RFE were added. As shown in Fig. S1 different CS films were fabricated. Abbreviations for the different film is as: C1 film (Control CS film), C2 film (CS film with 5 % MgO nanoparticles), C3 film (CS film with 5 % MgO nanoparticles and 5 % RFE), C4 film (CS film with 5 % MgO nanoparticles and 15 % RFE) and C5 film (CS film with 5 % MgO nanoparticles and 25 % RFE). The limit of concentration of RFE was chosen based on the preliminary screening study, in which it was observed that an equal or higher concentration than 30 % (*w/w*) caused difficulty in the film peeling-off process; in which film would break during the peeling off process and remained stuck to the surface of acrylic surface, suggesting degrading mechanical property.

### 2.4. Characterization of the CS films

An IR spectrometer (Bruker Corp., Alpha II FTIR, Germany) equipped with a platinum ATR diamond module was used to evaluate the CS films. OPUS version 6.5 was used for analysis after an average of 16 scans per sample were taken at a resolution of 4 $cm^{-1}$ in the 600–4000 $cm^{-1}$ range. Utilizing a thermal gravimetric analyzer (Perkin Elmer Inc., TGA-7 Pyris 1, USA), the thermal characteristics of the CS films were investigated. A 50 mg sample was heated at a ramp rate of 10 °C/min in a nitrogen environment, covering a temperature range of 30 °C to 600 °C.

XRD analysis, conducted with a Rigaku Corp., Miniflex instrument, Japan using Cu Kα radiation (1.54065 Å), assessed the crystal structure of CS films and MgO nanoparticles by acquiring spectra over the 4 to 70° (2θ) range at a scanning rate of 5°/min. The quantification of the crystallinity index (CI%) was carried out using the following formula (Schutz et al., 2023):

$$CI(\%) = \left(\frac{A_1}{A}\right) \times 100 \quad (1)$$

where, total area and area of the crystalline peak are denoted by A and $A_1$, respectively.

The ESEM (Field Electron and Ion Co., Quanta 200, Netherlands) was used to examine the sample's surface morphology. The sample's conductivity was enhanced with the application of a 10-s cycle of a gold coating that was 2–5 nm thick. The Energy-dispersive X-ray analysis (EDAX) was performed to determine the present of elements in the CS film. The EDAX was performed in field emission gun nano nova scanning electron microscope (FEI Ltd., Nova NanoSEM 450, USA) with EDAX operating at 15 kV. In accordance with the method described in Luo et al. (2022), the measurement of moisture content (MC), water solubility (WS), and water vapor permeability (WVP) was carried out (Luo et al., 2022).

The universal testing equipment (Instron Corp., Instron-5967, USA) was employed to assess the tensile strength (TS) and % elongation at the break of the CS films. The CS film was cut into a film specimen with sample size of 120 × 20 mm. The test was conducted at a crosshead speed of 5 mm per min, with an initial gauge length value of 60 mm. The measurement of the thickness of the CS film was carried out using the

thickness gauge meter (0.01 mm precision). The reported average thickness resulted from five random measurements across three film samples.

### 2.5. Light properties of the CS films

UV-spectra analysis was evaluated using a microplate reader (SPECTROstar nano, BMG labtech, Germany) in the range of 200–800 nm wavelength. Opacity and transparency of the CS films were assessed using the method described in the previous literature (Terzioğlu et al., 2021).

### 2.6. Contact angle measurement

A contact angle meter (Dataphysics Corp., OCA 15EC, USA) with SCA software was utilized to evaluate the contact of CS films. A precisely calibrated water droplet of 1 μL was dispensed using a microneedle over the surface of the CS film. Dynamic contact angle assessment was carried out for 1 min time period, with the measurement taken at every 10 s intervals to elucidate the wettability characteristic.

### 2.7. Antioxidant study of the CS films

DPPH and ABTS radical scavenging activities were measured using the modified method as depicted in Bhatia et al. (2024), in which 50 mg CS film sample was mixed with 1.9 mL of DPPH solution. The solution was vigorously shaken for 30 s, followed by keeping the solution in a dark environment for 30 min at room temperature. The absorbance was measured at 517 nm. ABTS radical scavenging activity was also measured, in which 30 mg CS film sample was mixed with 1.9 mL of $ABTS^{+}$ solution. The solution was kept in the darkroom for 30 min, and the absorbance was measured at 734 nm (Bhatia et al., 2024).

### 2.8. Overall migration (OM) study

OM study was carried out using the previously described method. The safety of the CS films was assessed in accordance with EU 10/2011 (Agustinelli et al., 2021). Safety of the CS films in the two food simulants, namely, 5 % ethanol and isooctane simulating aqueous and fatty food environment, respectively, were used. The 10 $dm^2$/L CS films were kept in a glass vial containing food simulant and kept at room temperature conditions for 10 days. After 10 days, the films were taken out, and the simulant liquid was evaporated in a heating oven. The residual obtained was measured gravimetrically in analytical balance (Mettler Toledo India Pvt. Ltd., ME54/A04, India, accuracy ± 0.0001 g).

### 2.9. Biodegradation study of the films

The CS films were subdivided into pieces measuring 2.5 × 2.5 $cm^2$, and initial weight was measured. Afterward, the CS films were positioned at a depth of 75 mm within containers filled with organic compost. To sustain the moisture content of 40–50 %, the appropriate amount of water was sprayed twice daily. At 5 days of time period, the films were taken out from the container and cleaned to remove dust. The final weight was noted, and %weight loss was evaluated (Hasan et al., 2020). To observe any physical alterations during the same time period, the CS films were subjected to photographic documentation.

### 2.10. Statistical analysis

The statistical analysis was carried out using Minitab 18 (Minitab LLC, Version 18, USA). The data were analyzed by one-way analysis of variance (ANOVA) and followed by tuckey's post hoc analysis. The values presented in the tables are shown with different alphabet representing statistically significant result with a significance level of $<0.05$. The measurements were done in triplicates, and the average value is reported.

## 3. Result and discussion

### 3.1. Total phenolics, total betacyanin, and total betaxanthin content

The total phenolic content was found to be 27.38 ± 0.67 mg per g of extract. Total betacyanin and total betaxanthin content was found to be 383.01 ± 10.16 mg per g of extract and 334.05 ± 3.07 mg per g of extract, respectively. The vacuum-dried powder of red dragon fruit peel was reported to have 175 mg/g of betacyanin content (Qin et al., 2020b). The betacyanin compounds present in red dragon fruit peels are phyllocactin, isophyllocactin, betanin, and isobetanin (Qin et al., 2020a). Indicaxanthin and its isomer are the major betaxanthin compounds present in dragon fruit peels (Huang et al., 2021).

### 3.2. FTIR analysis of the biofilms

FTIR serves as a crucial method for analyzing the intermolecular interactions between the CS, MgO nanoparticles, and RFE. FTIR spectra of C1, C2, C3, C4, and C5 films is shown in the Fig. S2 (a). The absorption band at 3272 $cm^{-1}$ represents the –OH stretching vibrations, and 2918 $cm^{-1}$ exhibits asymmetric as well as symmetric stretching of –NH bonds in the CS matrix. For the control film (C1 film), the bands at 3272 $cm^{-1}$ shifted to 3291 $cm^{-1}$ in C2 film, 3254 $cm^{-1}$ in C3 film, 3240 $cm^{-1}$ in C4 film, and 3233 $cm^{-1}$ in C5 film. This confirms the hydrogen bond interactions between CS, MgO nanoparticles, and RFE. The peak becomes less broad at stretching at 3272 $cm^{-1}$, showing that after adding RFE, the –OH bending involved in hydrogen bonding is disrupted or altered. (De Silva et al., 2017). Lan et al. (2021) also reported similar findings for intermolecular interactions between red apple pomace extracts, $TiO_2$ nanoparticles, and CS-based biofilm (Lan et al., 2021). The carbonyl bonds of amide groups (C=O-NHR) at 1647 $cm^{-1}$ in the C1 and C2 films changed to 1637 $cm^{-1}$ and 1659 $cm^{-1}$ for C3 and C4 films, respectively (Liu et al., 2021b). The amine group at 1569 $cm^{-1}$ for C1 film shifted to 1553 $cm^{-1}$ after adding MgO nanoparticles in C2 film; a similar peak of 1553 $cm^{-1}$ was observed for C3, C4, and C5 films. An increase in sharpness at 1569 $cm^{-1}$ indicates that MgO nanoparticles may promote the interaction within the secondary protein structure. A similar shift in amine groups of CS film was reported at 1576 $cm^{-1}$ with the addition of MgO nanoparticles (De Silva et al., 2017). The $CH_2$ group at 1416 $cm^{-1}$ for C1 film shifted to 1411 $cm^{-1}$ for C2 and C3 film. For C4 and C5 films, it shifted to 1407 $cm^{-1}$. The peak at 1411 $cm^{-1}$ becomes sharper, indicating that MgO nanoparticles could facilitate the interactions and bonding with the $CH_2$ group of CS. The asymmetric vibrations of C═O at 1151 $cm^{-1}$ for C1, C2, and C3 films shifted to 1118 $cm^{-1}$ for C4 and C5 films (De Silva et al., 2017). The vibrational frequency of C—O stretching in the glycosidic linkages at 1022 $cm^{-1}$ for C1 and C2 film shifted to 1024 $cm^{-1}$ and 1016 $cm^{-1}$ for C3 and C4 films, respectively (Yadav et al., 2021). Alterations in these functional groups may lead to physiochemical changes in the CS composite films.

### 3.3. XRD analysis of the CS biofilm

Fig. S2 (b) shows the diffractograms of the CS films, which were conducted in order to evaluate the crystalline structure and asses the compatibility between the CS polymer, MgO nanoparticles, and RFE. As reported in Table S2, the C1 film showed a small diffraction peak at 9.56° and a strong diffraction peak at 19.52°, assigned to crystal forms I and II, respectively (Abdel Aziz and Salama, 2018). With the addition of MgO nanoparticles (C2 film), no new peaks were observed; however, a significant increase in CI (%) was observed. With the addition of RFE (C3 film), the decrease in CI (%) was observed due to the disruption of the crystalline structure owing to the presence of polyphenolic compounds. Adding banana peel extract containing phenolic compounds to chitosan film led to decreased crystallinity (Zhang et al., 2020). However, with an

increase in the concentration of RFE, the C3 film and C4 film showed improvement in CI (%); this could be because of an increase in cross-linking and creation of new hydrogen bonds, the formation of regular domains, resulting in an increased crystallinity. Therefore, it can be noted that an increased concentration of RFE led to a marginal increase in TS values (from Table 1) accompanied by an increase in CI(%) for C3, C4, and C5 films. Luo et al. (2022) also reported an increase in crystallinity by adding MgO nanoparticles and tea polyphenols into the potato starch biofilms (Luo et al., 2022). It is imperative to avoid a unilateral assumption regarding the linear correlation between the tensile strength and crystallinity of the films, as elevation in crystallinity may not uniformly lead a proportional increase in the tensile strength of biofilms. Bhatia et al. (2024) reported that a significant decrease in TS was observed with an increased concentration of black pepper essential oil, but an increase in crystallinity was observed during the XRD analysis (Bhatia et al., 2024). An increase in the concentration of black soybean coat extract rich in anthocyanins into chitosan led to a significant rise in TS, but a decrease in crystallinity was reported (Wang et al., 2019b).

### 3.4. Thermal properties of the CS composite films

The thermal stability exhibited by CS films underscores their capacity to withstand decomposition at elevated temperatures, a fundamental attribute crucial to their operational integrity in diverse applications. TGA thermographs are depicted in Fig. S2 (c). Notably, all CS films demonstrated three stages of weight loss. The initial stage of degradation (30–130 °C) and second stage of degradation (130–240 °C) can be attributed to the vaporization of physically bound moisture and glycerol, respectively (Luo et al., 2022). The third stage of degradation (above 240 °C) is due to the thermal degradation of the CS matrix and RFE (Yong et al., 2019). Examination of the tabulated data (Table S3) reveals that the control film (C1) exhibits the least thermal stability. Notably, the inclusion of 5 % MgO nanoparticles (C2 film) resulted in a significant increase in the thermal stability of the CS film. For C2 film, $T_{d35}$ and $T_{d50}$ increased respectively to 544 and 324 °C, indicating 16.73 % and 4.5 % improvement, as compared to C1 film. This enhancement can be exclusively ascribed to the inherent thermal stability of the MgO nanoparticles (thermal degradation starts at 2800 °C). It is posited that the presence of MgO nanoparticles in the CS matrix impedes the diffusion of thermally degraded CS products. The findings align with that of De silva et al. (2017), where similar improvement in the thermal stability of the CS film was observed when 5 % MgO nanoparticles were added to the CS film forming solution (De Silva et al., 2017). Conversely, for the C3 film, adding 5 % RFE yields a reduction in $T_{d35}$ and $T_{d50}$ temperature values as compared to C2 film, albeit these values remain higher to those of the C1 film. This suggests that RFE at a low concentration (5 % in the current study) may disrupt the more compact network established by CS and MgO nanoparticles. However, increasing the concentration of RFE (to 15 % and 25 %) may have contributed to stabilizing the film matrix structure. Increasing the concentration of RFE led to the enhancement in the intermolecular interaction between phenolic compounds, betacyanin compounds, betaxanthin compounds, MgO nanoparticles and the CS film matrix. This can be supported by the evidence that increasing the concentration of RFE to 15 % (C4 film) and 25 % (C5 film), also increased $T_{d35}$ and $T_{d50}$ temperatures. The C5 film showed the highest $T_{d50}$ temperature of 342 °C, which is 10.32 % increment from C1 film. Liu et al. (2021a) reported the heightened thermal stability of CS film due to intermolecular interaction between $TiO_2$ nanoparticles, litchi peel extracts and CS (Liu et al., 2021a). Similar to our result, Yong et al. (2019) also observed that increasing the concentration of purple-fleshed sweet potato extract (5 to 15 wt%) in the CS film led to significant enhancement in thermal stability (Yong et al., 2019).

### 3.5. Light properties

The physical appearance of the CS films is depicted in Fig. S1. From Table 1, it can be observed that the opacity of the C1 film (1.89 ± 0.61 AU nm/mm) sharply increases with the addition of the MgO nanoparticles (7.96 ± 0.39 AU nm/mm). In the literature, the opacity value of pure CS is 1.58 ± 0.08 AU nm/mm, which is closer to our value (Liu et al., 2021a). Adding RFE further increases CS film's opacity in the range of 10.28 ± 0.66 to 11.56 ± 1.06 AU nm/mm. Wang et al. (2019a) also observed an increase in the opacity of CS/herba lophatheri extract (HLE) film with an increase in HLE concentration (Wang et al., 2019a). A similar trend is observed when measuring the transparency. For the control film, the transparency is 33.15 ± 2.83 %; this means that the control film allows more light to pass through it. Its value decreases to 24.75 ± 1.59 % for C2 film. Furthermore, the addition of RFE led to a further decrease in transparency. With increasing concentration of RFE (from 5 to 25 %), there seems to be a continuous trend of smooth decrease of % transparency from 21.00 ± 4.13 % to 17.57 ± 2.58 %. From Fig. S2 (d), it can be observed that the C1 film has the lowest UV-light barrier property (200–370 nm). This is because of the absence of UV-light absorbing functional groups in the CS matrix structure (Vejdan et al., 2016). The addition of MgO nanoparticles improves the UV-light barrier property, because MgO nanoparticles have UV-light absorption capacity (De Silva et al., 2017). A similar result is reported in the literature in which a significant increase in the UV-blocking ability of polyacrylonitrile nanofibrous mats was observed with the addition of MgO nanoparticles (Dadvar et al., 2011). The addition of RFE significantly improves the UV-light barrier property. This is because of the UV-light absorbing capacity of the polyphenolic compounds present in RFE. The exclusive interaction between MgO nanoparticles and phenolic compounds may have resulted in the inhibition of the light to pass through the film. Similar results are reported in the literature (Liu et al., 2021b; Zhang et al., 2019).

**Table 1**
Thickness, TS, elongation at break, WS, MC, WVP, film opacity, and transparency of the C1, C2, C3, C4, and C5 films.

| Sample | Thickness (mm) | TS (MPa) | Elongation at break (%) | WS (%) | MC (%) | WVP ($\times 10^{-10}$) g $m^{-1}$ $Pa^{-1}$ $s^{-1}$ | Film opacity (AU nm/mm) | Transparency (%) |
|---|---|---|---|---|---|---|---|---|
| C1 film | $0.0537 \pm 0.0068^{b}$ | $17.83 \pm 1.42^{a}$ | $124.37 \pm 1.63^{a}$ | $15.18 \pm 1.89^{c}$ | $16.25 \pm 0.01^{b}$ | $2.58 \pm 0.02^{b}$ | $1.89 \pm 0.61^{c}$ | $33.15 \pm 2.83^{a}$ |
| C2 film | $0.0540 \pm 0.0040^{b}$ | $12.68 \pm 0.76^{b}$ | $132.91 \pm 2.71^{a}$ | $27.27 \pm 1.07^{b}$ | $24.75 \pm 0.02^{a}$ | $2.75 \pm 0.03^{a}$ | $7.96 \pm 0.39^{b}$ | $24.75 \pm 1.59^{b}$ |
| C3 film | $0.0667 \pm 0.0030^{a}$ | $3.21 \pm 0.21^{c}$ | $112.93 \pm 0.77^{b}$ | $34.73 \pm 0.85^{a}$ | $24.37 \pm 0.05^{a}$ | $2.76 \pm 0.04^{ab}$ | $10.28 \pm 0.66^{ab}$ | $21 \pm 4.13^{bc}$ |
| C4 film | $0.0683 \pm 0.0032^{a}$ | $3.44 \pm 2.05^{c}$ | $108.60 \pm 3.42^{b}$ | $38.29 \pm 1.87^{a}$ | $25.64 \pm 0.03^{a}$ | $2.79 \pm 0.14^{ab}$ | $10.69 \pm 2.34^{ab}$ | $19.75 \pm 0.93^{bc}$ |
| C5 film | $0.0692 \pm 0.0013^{a}$ | $3.46 \pm 0.63^{c}$ | $108.47 \pm 2.95^{b}$ | $38.58 \pm 2.81^{a}$ | $26.62 \pm 0.05^{a}$ | $2.83 \pm 0.01^{a}$ | $11.56 \pm 1.06^{a}$ | $17.57 \pm 2.58^{c}$ |

The values presented in the tables are shown with different alphabets representing statistically significant result with a significance level of $<0.05$.

### 3.6. ESEM and EDAX analysis

ESEM analysis was carried out to assess the changes in the surface morphology with the addition of MgO nanoparticles and varying concentrations of RFE into the CS film matrix. The ESEM images of different CS biofilms are depicted in Fig. S3. The control film (C1 film) exhibited a smooth and homogenous surface without any voids and cracks. The C2 film exhibited clearly visible aggregates of MgO nanoparticles on the surface. De Silva et al. (2017) also reported the presence of aggregates of MgO nanoparticles in the CS film with 5 % and 10 % MgO nanoparticles in films during the SEM analysis (De Silva et al., 2017). For C3, C4, and C5 films, increasing the concentration of RFE led to enhancement in the surface roughness and irregularities of the polymeric matrix. This could suggest the interface and intermolecular interactions between MgO nanoparticles, RFE, and CS. Lan et al. (2021) reported an increase in surface roughness of CS film with $TiO_2$ nanoparticles and red apple pomace extract as compared to control CS film (Lan et al., 2021). CS film with ZnO nanoparticles with 0.4 % bamboo leaves antioxidant exhibited the highest surface roughness compared to CS film with ZnO nanoparticles (Liu et al., 2021b). Liu et al. (2021a) also reported similar observations for the CS film incorporated with litchi peel extract and $TiO_2$ nanoparticles (Liu et al., 2021a). Fig. S4 show the EDAX profile of the C1 and C2 films. For C1 film, the elements detected are C and O with 57.74 and 42.26 atom (%), respectively. While for C2 film, the presence of C, O and Mg was confirmed with 51.36, 46.46 and 2.18 atom (%) respectively. The decrease in carbon percentage and increase in oxygen percentage in the C2 film compared to C1 film is due to the presence of MgO nanoparticles. This confirms the successful incorporation of the MgO nanoparticles into the chitosan film matrix.

### 3.7. Mechanical properties and thickness of the CS films

The control film's TS (C1 film) was determined to be 17.83 ± 1.42 MPa, closely aligning with the literature-reported value of 17 MPa (Pereda et al., 2008). After introducing 5 % (*w*/w) MgO nanoparticles, the TS was reduced to 12.68 ± 0.76 MPa. In contrast, De Silva et al. (2017) noted an increase in TS by incorporating 5 % (w/w) MgO nanoparticles to 37.24 MPa by utilizing high molecular weight CS. The reason for the contradiction could be the use of medium molecular-weight CS in the current study. According to De Silva et al. (2017), the enhancement in mechanical property of the CS film was due to interfacial interaction between hydroxyl and amine group of the MgO nanoparticles by means of hydrogen bonding. Due to aggregation of MgO nanoparticles, researcher observed decrease in tensile strength at 10 % (*w/v*) MgO nanoparticle addition. The authors reported weaker interfacial interactions at higher MgO nanoparticles loading (De Silva et al., 2017). Alakayleh et al. (2016) reported that molecular weight of the chitosan is equivalent to that D-glucosamine repeating units, and in each repeating units, interactive amine groups ($-NH_2$) are bonded. The longer chain of chitosan has more $NH_2$ sites available (Alakayleh et al., 2016). Amino (-$NH_2$) and hydroxyl (-OH) groups are present in the CS, constituting potential binding sites for MgO nanoparticles. Higher molecular weight CS may provide more binding site because of more number of repeating units. Medium molecular weight CS may provide a lower number of binding sites, resulting in MgO nanoparticles agglomeration. This may result in a decrease in TS and an increase in % elongation at break. Consistent with the literature, Liu et al. (2021b) reported a decline in TS of CS film with the addition of ZnO nanoparticles. Also, reduction in TS was reported with the addition of antioxidant of bamboo leaves (Liu et al., 2021b). In our investigation, adding RFE resulted in a decline in TS within the range of 3.21 ± 0.21 to 3.46 ± 0.63 MPa. The presence of bioactive compounds, including polyphenolic, betacyanin, and betaxanthin compounds in RFE, coupled with MgO nanoparticles, likely induced heightened agglomeration in the CS network.

This phenomenon may cause disruption in the crystalline structure, thereby restricting the inter-chain interaction of the CS and concomitantly resulting in a decline in TS of C3, C4, and C5 films. Liu et al. (2021b) postulated that polyphenolic compounds might disrupt the polymer chain interactions within the CS network, resulting in a decrease in TS (Liu et al., 2021b). From Table 1, it can be observed that the C3, C4, and C5 films have lower values of % elongation at break as compared to C1 and C2 films. This could be because of the disruption in intra- and intermolecular hydrogen bonds of CS chains by the polyphenolic compounds. Consistent with this, Bi et al. (2019) reported a reduction in % elongation at the break of CS film by adding proanthocyanidins (Bi et al., 2019). Rubilar et al. (2013) noted an analogous decrease in TS and % elongation at break of CS film with different concentrations of carvacrol and grape seed extracts (Rubilar et al., 2013). The C5 film exhibited a significant increase in thickness up to 0.0692 ± 0.0013 mm because of the increase in the solid contents (Riahi et al., 2021). Similar observations are also reported in the literature (Liu et al., 2021b; Yong et al., 2019).

### 3.8. Dynamic contact angle (°) measurement

The dynamic contact angle of distilled water over different CS composite films was carried out to evaluate the surface hydrophobicity. As shown in Fig. S5, the initial contact angle was measured, followed by measurement after 10 s interval for the total time period of 1 min. For C1 film, the contact angle changed from 101.55 ± 0.91° to 86.4 ± 1.13°. In the literature, the pure CS film is reported to have a static contact angle of 105° (Martins et al., 2012). With the addition of MgO nanoparticles, the contact decreased to 60.75 ± 4.59°. This could be because of the agglomeration of MgO nanoparticles. MgO nanoparticles are inherently hydrophilic in nature. Also, the surface area occupied by CS-incorporated MgO nanoparticles may be higher than that of the pure CS film. This may lead to a higher number of hydrophilic groups interacting with water and decreasing the contact angle. In the literature, the contact angle of the CS film incorporated with Ag nanoparticles was less than that of the pure CS film (Raghavendra et al., 2016). The dynamic contact angle of the C3, C4, and C5 film was less than that of the C1 film and decreased with time. Also, after 40 s time, it was not possible to measure the contact angle further, because of the swelling issue of the films. The decrease in the contact angle is due to the addition of hydrophilic extract. The betacyanin and betaxanthin compounds are water soluble and, therefore, hydrophilic in nature. Kaya et al. (2018) also reported decreased contact angle values of the CS films incorporated with Berberis Crataegina's fruit extract. The researcher ascribed this phenomenon to an increase in the chain mobility of CS matrix and the hydrophilic nature of the extract (Kaya et al., 2018). Augusto et al. (2018) also reported a significant decrease in contact angle up to 21.80° by incorporating *Codium Tomentosum* extract into CS film and correlated the high moisture content with low contact angle values. The author attributed this phenomenon to the presence of polar compounds in the extract. Because solutions with high surface coating capacity are more likely to spread easily and wet surfaces well, using these solutions on food can be effective. (Augusto et al., 2018).

### 3.9. Moisture content and water solubility

MC for the control film was found to be 16.25 ± 0.01 %. The MC increased with the addition of MgO nanoparticles to 24.75 ± 0.02 %. Furthermore, increasing the concentration of RFE led to a marginal increment in the MC in the range of 24.37 ± 0.05 to 26.62 ± 0.05 %. MC is correlated to the volumetric void space filled by water molecules within the network of the CS matrix structure (Pereda et al., 2011). The hydrophilic nature of RFE is responsible for the rise in MC value. Similarly, the enhancement in MC values was reported with the increasing concentration of grape seed extract to the CS films, and the author attributed the hydrophilic nature of grape seed extract to the observed phenomenon (Rubilar et al., 2013). WS represents the response

of the biofilm in the presence of the aqueous environment and is directly correlated to hydrophilicity of the CS films. For C1 film, the value of WS is 15.18 ± 1.89 %. A similar value of WS for the control CS film of 16.97 % is reported in the literature (Bi et al., 2019). The high water affinity of the hydrophilic group and glycerol present in the C1 film is responsible for the high WS value (Rubilar et al., 2013). For C2 film, the WS value increased to 27.17 ± 1.07 %. For C2 film, the MgO nanoparticles aggregation takes place, and the areas around the aggregates could have a slightly less ordered or more amorphous character due to nanoparticle interaction. These less ordered regions might be more accessible to water molecules compare to highly crystalline regions, potentially contributing to increase in WS of the C2 film. For C3, C4, and C5 films, the value of WS increased but remained in the range of 34.73 ± 0.85 to 38.58 ± 2.81 %. Increasing the concentration of honeysuckle flower extract from 0 to 30 wt% in the CS film also increased the WS values from 23.52 % to 31.57 %, because of the hydrophilic nature of honeysuckle flower extract (Wang et al., 2017). The data from dynamic contact angles are in line with the observations of the WS and MC. In some cases, elevated water solubility becomes a favorable property, especially in the case when the film or the coating are meant to be consumed along with the food material. High WS values are also favorable to enhancing the biodegradability of the biofilms. Conversely, in the packaging application of aqueous food materials, low water solubility is desirable (Rubilar et al., 2013). But, in the case of fatty food packaging, the behavior of biofilms towards fatty food simulants becomes an essential factor.

### 3.10. Water vapor permeability (WVP) of the CS films

The values of WVP for different CS films are shown in Table 1. The WVP represents the ability of the CS films to provide a barrier against water vapor traveling through the film. The C1 film had a WVP value of $2.58 \times 10^{-10}$ g $m^{-1}$ $Pa^{-1}$ $S^{-1}$, similar to the literature value of $2.68 \times 10^{-10}$ g $m^{-1}$ $Pa^{-1}$ $S^{-1}$ (Marinho et al., 2022). For C2 film, the value of the WVP increased by 6.58 %. Even though there is an increase in crystallinity index (as per XRD analysis) for C2 film, the aggregation of MgO nanoparticles is observed, which is evident from the TS data (discussed in the mechanical property section). The aggregated nanoparticles may increase the value of WVP (Liu et al., 2021b; Riahi et al., 2021). Uneven dispersion of MgO nanoparticles can create gaps or imperfections within the film matrix. These gaps could act as pathways for water molecules, facilitating faster diffusion and higher permeability. This result in increase in WVP of C2 film. This is also supported by the increase in MC% and WS% of the C2 film compared to the C1 film. With increasing concentration of RFE in the CS, the values of WVP increased to 6.97 %, 8.13 %, and 9.68 % for C3 film, C4 film, and C5 film, respectively, compared to C1 film. This may be correlated to an increase in the value of WS and MC. The RFE extract is made using the extraction by water as a solvent, which means the RFE extract powder and, thus, the bioactive compounds obtained at the end of the process are water soluble in nature. For example, betalain pigment is water soluble and has higher water affinity. This may contribute to the increase in WVP values. A similar trend in literature is reported (Bi et al., 2019; Liu et al., 2021b).

### 3.11. Antioxidant activity of the CS films

The DPPH and ABTS radical scavenging antioxidant activity are shown in Table 2. The antioxidant activity represents the ability of CS films to prevent the oxidation of the food material, which results in the maintenance of the quality of the food. For C1 film, the %DPPH and % ABTS radical scavenging activity was observed to be 5.51 ± 1.57 % and 32.86 ± 1.68 %, respectively. This is attributed to the interaction between leftover amino groups in CS and free radicals, generating stable macromolecular radicals and ammonium groups (Zheng et al., 2019). Liu et al. (2021b) reported 14.33 % DPPH and 31.99 % ABTS radical scavenging activity of the control CS film (Liu et al., 2021b). For C2 film, the %DPPH and %ABTS radical scavenging activity increased to 25.41 ± 1.07 % and 40.63 ± 1.65 % respectively. This aligns with the prior studies showing enhanced antioxidant activity of CS film with the addition of metal oxide nanoparticles (Liu et al., 2021b; Zhang et al., 2019). For C3, C4, and C5 films, the %DPPH radical scavenging activity significantly increased to 27.72 ± 1.34 %, 31.23 ± 1.42 % and 37.64 ± 0.83 % respectively. The corresponding % ABTS radical scavenging activity increase was observed to be 44.46 ± 1.63 %, 54.65 ± 1.44 % and 62.83 ± 1.51 %.

**Table 2**
DPPH radical scavenging activity (%), ABTS radical scavenging activity, OM values for 5 % ethanol and isooctane of C1, C2, C3, C4, and C5 films.

| Sample | DPPH antioxidant activity (%) | ABTS antioxidant activity (%) | OM values for 5 % ethanol simulant (mg/kg of simulant) | OM values for isooctane simulant (mg/kg of simulant) |
|---|---|---|---|---|
| C1 film | $5.51 \pm 1.57^{e}$ | $32.86 \pm 1.68^{d}$ | $36.63 \pm 7.93^{d}$ | n.d |
| C2 film | $25.41 \pm 1.07^{d}$ | $40.63 \pm 1.65^{c}$ | $306.78 \pm 34.57^{c}$ | n.d |
| C3 film | $27.72 \pm 1.34^{c}$ | $44.46 \pm 1.63^{b}$ | $828.75 \pm 41.96^{b}$ | n.d |
| C4 film | $31.23 \pm 1.42^{b}$ | $54.65 \pm 1.44^{a}$ | $1208.79 \pm 27.47^{a}$ | n.d |
| C5 film | $37.64 \pm 0.83^{a}$ | $62.83 \pm 1.51^{a}$ | $1254.58 \pm 20.98^{a}$ | n.d |

The values presented in the tables are shown with different alphabets representing statistically significant result with a significance level <0.05. *n.d: not detected.

The polyphenolic and betalain compounds present in RFE, along with MgO nanoparticles, have an inherent capacity to capture free radicals effectively and inhibit the generation of reactive oxygen species. This implies that the synergistic effect of RFE and MgO nanoparticles resulted in a significant increase in the antioxidant activity of the CS films. Luo et al. (2022) reported a significant increase in the DPPH radical scavenging antioxidant activity of the potato starch film with the addition of tea polyphenol and MgO nanoparticles (Luo et al., 2022). Up to 72.85 % DPPH radical scavenging of CS film incorporated with $TiO_2$ nanoparticles and black plum peel extract was reported (Zhang et al., 2019). Up to 69.82 % DPPH radical scavenging activity and 87.93 % ABTS radical scavenging activity was reported for CS film incorporated with ZnO nanoparticles and bamboo leaves (Liu et al., 2021b).

### 3.12. Overall migration study of the CS films

Overall migration study of all CS films was carried out using the two simulants: (a) 5 % ethanol and (b) isooctane at room temperature. As shown in Table 2, C1 films exhibited a value of OM less than the permissible limit of 60 mg/kg, while the value of OM increased significantly for C2, C3, C4, and C5. The increase in OM value directly correlates with WS values reported in the previous section. The higher water affinity of RFE extract is directly responsible for the observed phenomena. On the contrary, the OM value (not detected) in the isooctane simulant was found to be lower (as shown in Table 2) than the permissible limit, making the CS films safe to use in fatty food environments in accordance with commission regulation (EU) No. 10/2011. There is a lack of significant diffusion of isooctane within the polar polymer matrix. This led to sustained closure of the polymer structure. Therefore, it can be inferred that when the CS film comes in contact with non-polar systems, such as isooctane, the polar polymeric system remains structurally sound over the period of storage (Sánchez-González et al., 2011). Similar results were found for soy protein isolate film with red grape extract (Agustinelli et al., 2021), poly(butylene adipate-*co*-terephthalate), and electrospun CS nanofibers based biofilm (Díez-Pascual and Díez-Vicente, 2015).

### 3.13. Biodegradation study

The soil degradation test was carried out for different CS films. It can be observed from the Table S4 that after 5 days of the degradation time, mass loss (%) was lowest for C1 film at 8.62 %, while C2, C3, C4, and C5 film was in the range of 20.61–34.16 %. However, after day 10, all the films were found to be completely degraded. From Fig. 1, it can be observed that after 5 days, the films showed a profound transformation in visual appearances, characterized by the loss of their initial appearance and structural integrity. This degradation manifested as rougher and eroded surfaces exhibiting perforations. These observations provide compelling evidence for the substantial biodegradation by the indigenous soil microbial community. Deshmukh et al. (2021) investigated the biodegradability of CS-based biofilms incorporated with varying concentrations (0–25 wt%) of defatted Chlorella biomass through a 60-day soil burial test. Notably, the researcher observed extensive biodegradation, exceeding 50 %, highlighting the significant potential of these biofilms as sustainable materials (Deshmukh et al., 2021). Oberlintner et al. (2021) reported complete biodegradation of CS film enriched with Quercus polyphenol extract into industrial compost, vineyard soil, and garden soil within 10 days of time period (Oberlintner et al., 2021). It is essential to highlight that the commencement of degradation in all CS-based biofilm specimens predates the expiration of the 180-day period. This threshold is stipulated for attaining biodegradability, necessitating a minimum of 90 % disintegration of plastic material within this specified timeframe (ASTM D6400–12). Based on soil test results, all the investigated CS based biofilms demonstrated complete biodegradability. Notably, this contrasts with low-density polyethylene (LDPE) films, which exhibit a degradation rate of only 0.2 wt% per decade (Albertsson, 1980). This characteristic translates to two key advantages: firstly, a minimal environmental footprint on terrestrial ecosystem through complete biodegradation, leaving no residual traces. Secondly, a significant decrease in waste creation across both the food processing industry and the plastic waste stream.

### 3.14. Comparison with literature

In the current work, % DPPH and %ABTS radical scavenging activity increased up to 37.64 ± 0.83 % and 62.83 ± 1.51 % respectively for C5 film. While tensile strength decreased to 12.68 ± 0.76, 3.21 ± 0.21, 3.44 ± 2.05, and 3.46 ± 0.63 MPa for C1, C2, C3, C4 and C5 film. As shown in Fig. S2 (d), the C2, C3, C4 and C5 films does not any light transmittance in the UV-light range of 100 to 380 nm, showing excellent UV-light resistance property. As shown in Table 3, addition of MgO nanoparticles and tea polyphenols into the potato starch film led to increase in the UV-resistance, antioxidant activity and tensile strength. The improvement are primarily because of interfacial adhesion of the polyphenols and nanoparticles (Luo et al., 2022). CS was incorporated with 1.25 % (*w/v*) of ZnO nanoparticles and 0.1 % to 0.4 % (w/v) bamboo leaves, showed that with addition of only ZnO nanoparticles marginally decreased, while increasing concentration of bamboo leaves completely blocked the light in the UV range. This was due to significant UV-light was absorbed by the flavonoids present in the bamboo leaves. The flavonoids have strong antioxidant activity, because of which the

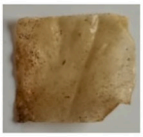
| Sample | Day 1 | Day 5 | Day 10 |
|---|---|---|---|
| C1 film |  |  | - |
| C2 film |  |  | - |
| C3 film |  |  | - |
| C4 film |  |  | - |
| C5 film |  |  | - |

**Fig. 1.** Physical appearance of different CS films undergoing soil degradation for day 1, day 5, and day 10.

**Table 3**
Comparison with the literature.

| Film details | Antioxidant activity | Mechanical strength | UV-resistance | Reference |
|---|---|---|---|---|
| CS film incorporated with MgO nanoparticles and RFE | 37.64 ± 0.83 % DPPH activity and 62.83 ± 1.51 % ABTS activity for C5 film | 12.68 ± 0.76, 3.21 ± 0.21, 3.44 ± 2.05, and 3.46 ± 0.63 MPa for C1, C2, C3, C4 and C5 film. | No light transmittance for C2, C3, C4 and C5 film in the UV-range indicating superior UV-resistance. | Current work |
| Potato starch film incorporated with tea polyphenol and MgO nanoparticles | Increase in DPPH activity up to 6.25 ± 0.40 mg gallic acid equivalent(GAE)/g | At 0.6 % tea polyphenol and 1.5 % MgO nanoparticles, highest tensile strength was 24.05 ± 0.77 MPa | Decreasing light transmittance with addition of MgO nanoparticles and tea polyphenols | (Luo et al., 2022) |
| CS film incorporated with ZnO nanoparticles and Bamboo leaves | 69.82 % and 77.89 % increase in %DPPH and %ABTS activity | TS decreased from 81.44 MPa to 12.43 MPa after addition of ZnO nanoparticles and Bamboo leaves | Significant decrease in light transmission for films incorporated with bamboo leaves and ZnO nanoparticles | (Liu et al., 2021b) |
| CS film incorporated with silver nanoparticles and purple corn extract | Increase in DPPH activity to >50 % | TS increased to 36.60 ± 1.95 | Decreasing trend in light transmission in the UV-light region | (Qin et al., 2019) |
| CS film with TiO2 nanoparticles and plum peel extract | DPPH radical scavenging activity increased up to 72.85 % | TS increased to 23.98 ± 0.13 MPa. | Decreasing trend in light transmission in the UV-light region | (Zhang et al., 2019) |

DPPH and ABTS radical scavenging activity increased. However, the ZnO nanoparticles and bamboo leaves created more aggregates and decreased the tensile strength from 81.44 MPa (for control film) to 12.43 MPa (for film incorporated with 1.25 % ZnO nanoparticles and 0.4 % bamboo leaves) (Liu et al., 2021b). In CS, with addition of 2 % Ag nanoparticles and 2 % purple corn extract, decreasing trend in light transmission in the UV-light region was reported. % DPPH radical scavenging activity also increased because of the Ag nanoparticle (by accepting or denoting electrons) and puple corn extract (because of presence of anthocyanins). Hydrogen bonding purple corn extract and CS increased the TS of the film (Qin et al., 2019). The addition of the purple corn extract (0.8 g) and TiO2 nanoparticles (0.8 g) into the CS (3.2 g) led to decreasing light transmittance in the UV-light region and increasing antioxidant activity because of the presence of the phenolic compound in the purple corn extract. Intermolecular interactions between TiO2 nanoparticles, extract, CS, and glycerol led to significant increase in TS to 23.98 MPa. (Zhang et al., 2019).

## 4. Conclusions

RFE and MgO nanoparticles were successfully incorporated in the CS-based biofilm. Hydrogen bonds interactions between RFE, CS and MgO nanoparticles were confirmed in the FTIR analysis. ESEM images revealed the aggregates of MgO nanoparticles, with increasing concentration of RFE contributing to a rougher film surface and EDAX analysis confirmed the presence of elements such as, Mg, O and C atoms in the C2 film. Among the films examined, the C5 film demonstrated the highest thermal stability, thickness (0.0692 mm), film opacity (11.56 AU nm/mm), water solubility (38.58 %), moisture content (26.62 %), and water vapor permeability ($2.83 \times 10^{-10}$ g $m^{-1}$ $Pa^{-1}$ $s^{-1}$). The combined effect of MgO nanoparticles and RFE led to a notable increase in DPPH and ABTS radical scavenging activity of the C5 film to 37.63 % and 62.83 %, respectively, compared to the C1 film. TS value decreased for C2, C3, C4 and C5 films, measuring 12.68, 3.21, 3.44, 3.46 MPa, respectively, although the UV-blocking property notably enhanced. The OM values for all CS films were below 60 mg/kg of simulant for fatty food. However, the OM values exceeded the permissible limit for an aqueous food environment, measuring 36.63, 306.78, 828.75, 1208.79, and 1254.58 mg/kg of simulant for C1, C2, C3, C4, and C5 respectively. Notably, all CS films completely degraded in the soil burial test within 5 days. These findings indicate that the developed CS biofilms, incorporating MgO nanoparticles and RFE, hold great promise as environmentally friendly and functional packaging materials for potential food applications.

## CRediT authorship contribution statement

**Rushikesh A. Prajapati:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Girirajsinh C. Jadeja:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Funding acquisition, Data curation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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*Ethical approval*

Not applicable. No experiments were conducted on humans or animals as part of this work.

## Appendix A. Supplementary data

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